Workflow System Design: NOC Incident Escalation Automation

Objective

Design an automated workflow that receives critical incident tickets from the OWS monitoring system, identifies the responsible field engineer from the OM Database, and immediately escalates the incident through Email and WhatsApp while monitoring for acknowledgement.

Business Process

Telecommunication NOC Incident Escalation

Workflow Components

Trigger

- A new **Critical Incident Ticket** is generated by the OWS Monitoring Tool.
- The AI assistant detects the incoming ticket automatically.

Workflow Steps

Step 1 – Receive Incident Ticket

- Monitor the OWS system or a dedicated email inbox continuously.
- Detect a newly generated critical incident ticket.

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Step 2 – Extract Ticket Information

- Read the ticket details.
- Extract:
    - Incident ID
    - Site ID
    - Site Name
    - Alarm Type
    - Severity

- Time Generated

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Step 3 – Validate Incident

- Confirm that the alarm is a Critical Site Down incident.
- If the ticket is not critical, follow the standard escalation process.

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Step 4 – Search OM Database

- Use the extracted **Site ID**.
- Search the uploaded OM Database.
- Retrieve:
  - Assigned Field Engineer
  - Email Address
  - WhatsApp Number
  - Region (if available)

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Step 5 – Generate Escalation Message

- Automatically create an escalation message containing:
  - Incident ID
  - Site ID
  - Site Name
  - Alarm Type
  - Time Raised
  - Priority Level

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Step 6 – Notify Field Engineer

- Send:

- Email
- WhatsApp Message

Both notifications are sent simultaneously.

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Step 7 – Start Acknowledgement Timer

- Start a **5-minute countdown**.

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Decision Point

Has the engineer acknowledged?

Yes

- Update ticket status to **Acknowledged**.
- Record acknowledgement time.
- Continue monitoring until resolution.

No

- Send an automatic reminder.
- Wait another 5 minutes.
- If there is still no acknowledgement, escalate to the Supervisor or Regional Engineer.

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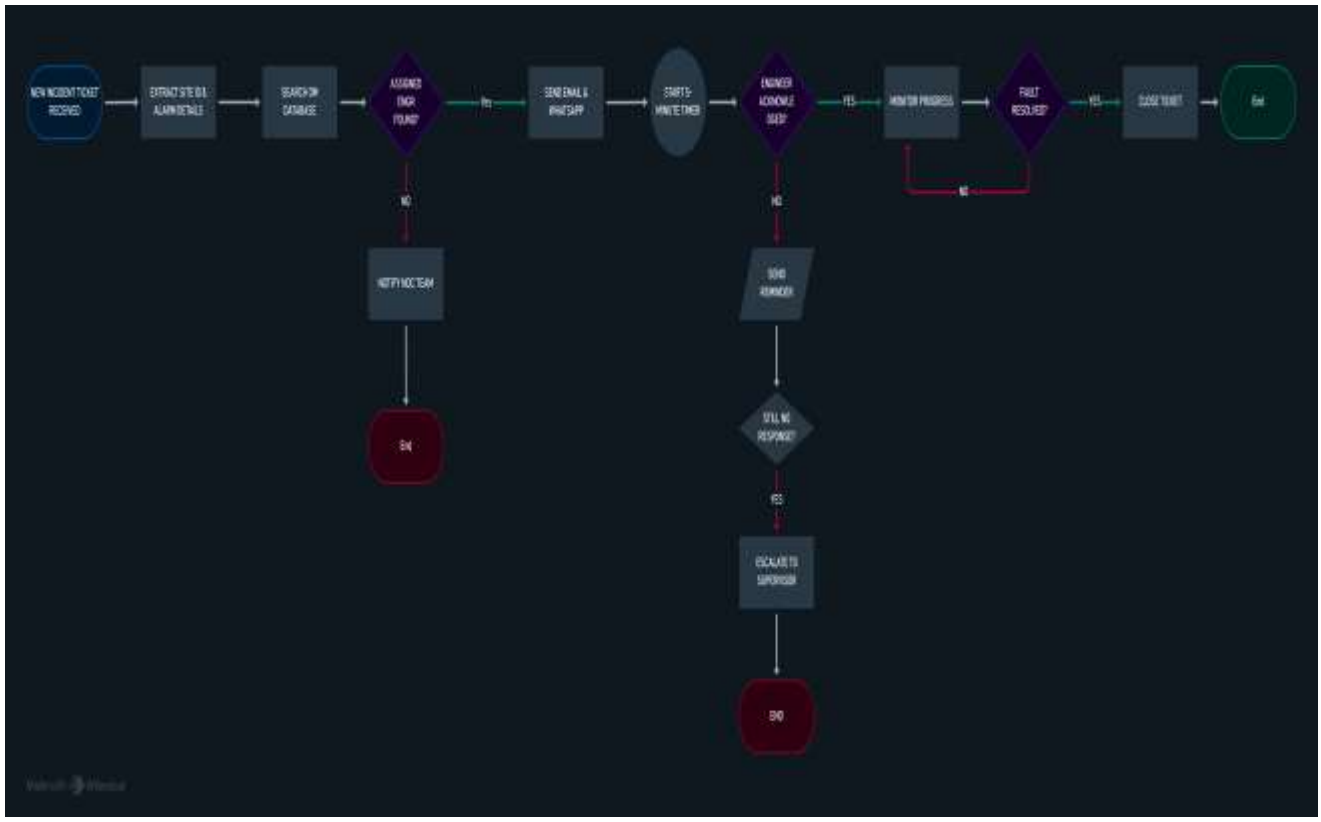
Step 8 – Close Incident

- When the engineer confirms the fault has been resolved:
  - Update the ticket.
  - Record resolution time.
  - Store the escalation history for reporting.

Workflow Logic

Trigger	Action	Logic
New incident ticket received	Extract Site ID	Read ticket details automatically
Site ID extracted	Search OM Database	Match Site ID to assigned Field Engineer
Engineer found	Send Email & WhatsApp	Immediate escalation
Wait 5 minutes	Check acknowledgement	If acknowledged, continue monitoring
No acknowledgement	Send reminder	Repeat escalation
Still no acknowledgement	Escalate to Supervisor	Follow escalation hierarchy
Fault resolved	Close ticket	Save incident log

Simple Workflow Diagram



https://whimsical.com/toluwanimi-akinyemi/Q2ikNoe7DPegzTGm6PxQdZ

Practical Relevance

This workflow aligns with the assignment requirements by clearly defining the **business process**, **trigger**, **actions**, **decision logic**, and **visual workflow structure**.

In a real telecom Network Operations Center (NOC), this automation would:

- Reduce Mean Time to Acknowledge (MTTA).
- Improve Mean Time to Repair (MTTR).
- Eliminate manual engineer lookup from the OM Database.
- Ensure every critical alarm is escalated consistently and without delay.
- Maintain a complete audit trail of notifications, acknowledgements, and resolutions.

This workflow is also well suited for implementation in automation platforms such as **n8n**, **Microsoft Power Automate**, or **UiPath**, with Python or AI services handling ticket parsing, database lookups, and intelligent escalation logic.